Chapter 6

EQUILIBRIUM

CONCEPT OF EQUILIBRIUM

Equilibrium is a situation which indicates balance between two opposite forces. A point where two curves moving in opposite direction intersect each other is known as "equilibrium point" e.g. Equilibrium of National Income where : Aggregate supply = Aggregate demand OR $y = C + I$ OR $S = I$.

Market equilibrium where DD (demand curve) intersects Ss (supply curve) as DD is negatively sloped while SS is positively sloped.

DEMAND AND SUPPLY EQUILIBRIUM OR MARKET EQUILIRBIUM

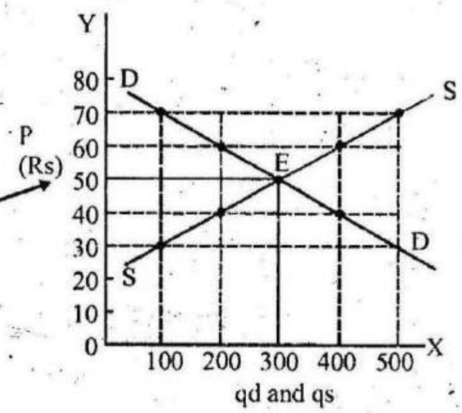
As $qd = f(p)$ and it is a decreasing function thus demand curve (DD) is negatively slopped moving downwardly from left to right.

On the other hand $qs = f(p)$ and it is an increasing function. Hence supply curve (SS) is positively sloped moving upward from left to right. Point of intersection of the both curves "E" shows that $qd = qs$ (equilibrium quantity). The price at which

quantity demanded = quantity supplied is known as "equilibrium price". This is the situation known as "Market equilibrium". Under perfect competition market price is determined at the point where $qd = qs$.

It is also explained with the help of a schedule and diagram taking in view the prices. Demand and supply of ghee (p,qd, qs) in the market.

P(Rs)	qd(K.G)	qs(K.G)
70	100	500
60	200	400
50	300	300
40	400	200
30	500	100



Explanation:

Keeping in view various combinations of p and qd in the above schedule. DD (demand Curve) has been drawn which is negatively sloped. Similarly Ss (supply curve) has been prepared with respect to different combinations of p and qs in the schedule. SS is positively sloped and moves upward left to right. As DD and Ss move in opposite direction thus they intersect each other at point E in the diagram. E is the equilibrium point showing p = (Rs.50) as equilibrium price at which qd = qs = $\boxed{300}$ equilibrium quantity.

EQUILIBRIUM IN VIEW OF DEMAND AND SUPPLY EQUATIONS: (MATHEMATICAL AND GRAPHIC APPROACH)

Example I :

(i) qd = 10 – 2P (demand function)

(ii) qs = 4 + p (supply function)

In equilibrium condition qd = qs

Find equilibrium price and equilibrium quantity in view of above mentioned demand and supply functions?

Solution:

(i) qd = 10 – 2p Deducting equation No. (ii) from No:- (i)

(ii) – qs = – 4 ± p

$$0 = 6 - 3p$$

$$3p = 6$$

$$p = \frac{6}{3} = \boxed{2} \text{ Equilibrium price.}$$

Putting value of p = 2 in demand function

qd = 10 – 2(2) = 10 – 4 = $\boxed{6}$ Equ. Qd

Now Putting value of p = 2 in supply function

qs = 4 + p = 4 + 2= $\boxed{6}$ Equ. qs

To complete the schedule we put Equ. Price in between assumed less and more values e.g. 1, 2, 3, 4

Putting these values one by one in demand function to find out qd.

(i) qd = 10 – 2p = 10 – 2(1) = 10 – 2 = $\boxed{8}$ At p = 1

(ii) qd = 10 – 2p = 10 – 2(2) = 10 – 4 = $\boxed{6}$ At p = 2

(iii) qd = 10 – 2p = 10 – 2(3) = 10 – 6 = $\boxed{4}$ At p = 3

(iv) qd = 10 – 2p = 10 – 2(4) = 10 – 8 = $\boxed{2}$ At p = 4

Now putting these values of p in supply function

(i) qs = 4 + p = 4 + 1 = $\boxed{5}$ At p = 1

(ii) qs = 4 + p = 4 + 2 = $\boxed{6}$ At p = 2

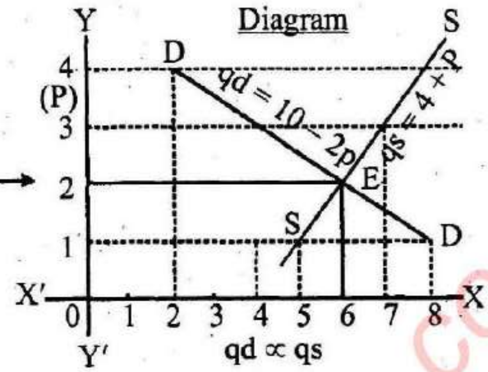
(iii) qs = 4 + p = 4 + 3 = $\boxed{7}$ At p = 3

(iv) qs = 4 + p = 4 + 4 = $\boxed{8}$ At p = 4

Putting values of p, qd and qs in the following scheduled and preparing diagram.

Schedule

P	1	2	3	4
qd	8	6	4	2
qs	5	6	7	8



Explanation:

As qd = qs = 6 and equ. P = 2 in the schedule.

Thus DD demand function and SS supply function intersect each other at point E in the diagram.

Point E also shows that equ. P = $\boxed{2}$ and equ. Quantity = $\boxed{6}$

Example 2:

(i) $q + p = 5$

(ii) $2p - q = 5.5$

(a) Find demand and supply function.

(b) Find equ. Price and equ. Quantity.

(c) Give graphic representation also.

Solution

Standard form of the given equations

(i) $q = 5 - p$ (demand function as co-efficient of p is –tive)

(ii) $-q = 5.5 - 2p$

(iii) $q = -5.5 + 2p$ (supply function as co-efficient of p is +tive)

We solve these equations to find equilibrium price.

$$
\begin{array}{lrcl}
\text{(i)} & q & = & 5 - p \\
\text{(ii)} & -q & = & \mp 5.5 \pm 2p \\
\hline
 & 0 & = & 10.5 - 3p \\
 & 3p & = & 10.5 \\
 & p & = & \dfrac{10.5}{3} = \boxed{3.5} \text{ Equ. Price}
\end{array}
$$

To find equ. qd we put equ. P (3.5) in Equation No. (i)

(i) $q = 5 - p = 5 - 3.5 = \boxed{1.5}$ Putting values of p(3.5) in No.(ii)

(ii) $q = -5.5 + 2(3.5) = -5.5 + 7 = \boxed{1.5}$

Thus qd = qs = $\boxed{1.5}$ Equilibrium quantity.

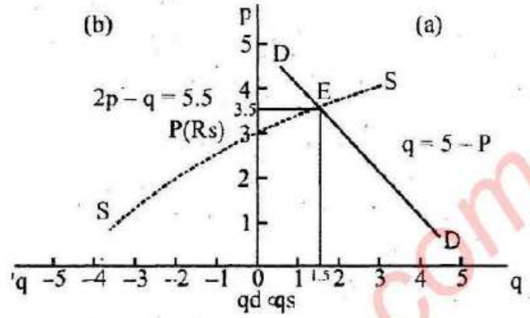
Now schedule and diagram

P	1	2	3	3.5	4
q	4	3	2	1.5	1

$q = 5 - p$

p	1	2	3	3.5	4
q	−3.5	−1.5	.5	1.5	2.5

$2p - q = 5.5$



Equ. Price = 3.5 and Equ. Quantity = 1.5 is shown in the schedule. Taking assumed p as 1,2,3,4 values of qd and qs have been entered against every value of p.

Keeping in view these combinations of p and qd (DD) demand curve and with the values of p and qs (SS) supply curve are drawn in the diagram. Both these curves intersect each other at point E, which is the equilibrium point. It shows Equ. q = 1.5 and equ. P = 3.5 as mentioned in the schedule.

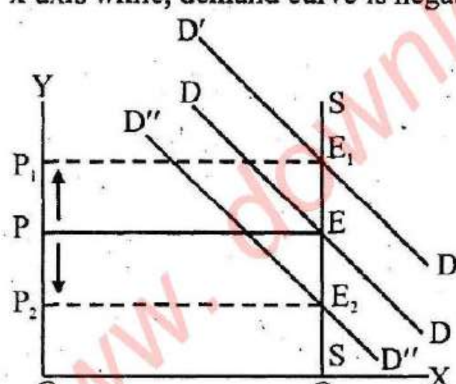
EFFECTS OF CHANGES IN DEMAND AND SUPPLY UPON EQU. PRICE AND EQU. QUANTITY

Any change in demand, supply or in both forces will indicate new equilibrium point showing a change in price, quantity or in both of them.

It can be explained with the help of graph under following conditions.

1. FIXED SUPPLY BUT AN INCREASE OR DECREASE IN DEMAND

In day to day market supply is fixed. Hence, Ss (supply curve) remains vertical upon x-axis while, demand curve is negatively sloped shown as under.



Explanation:

DD (demand curve) and SS (supply curve) intersect each other at point E.

Here OP is the equilibrium price and OQ equilibrium quantity.

(i) Due to increase in demand D′D′ shifts upward and new equ. Point is E_1.

(ii) When demand decreases D″D″ shifts downwards the original DD. New equilibrium point becomes as E_2.

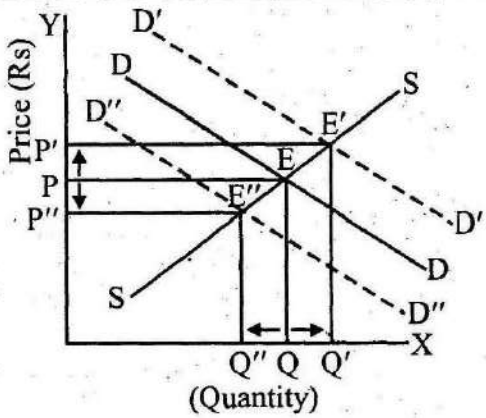
Effects:

(a) Quantity remains unchanged.

(b) (i) With the increase in demand, price moves upward from p to p_1

(ii) Due to decrease in demand, price falls from p to p_2.

2. SUPPLY UNCHANGED BUT RISE AND FALL INDEMAND



Explanation:

DD and SS intersect each other at point E. which indicates op as equ. P and OQ equ. Quantity.

(a) SS remaining the same when DD rises it shifts upward as D'D'. New equ. Poiont is E'.

(b) DD shifts downward as D''D'' with the fall in demand E" is the new equilibrium point.

Effects (a)

(i) Equ. P rises to p' and becomes op'.

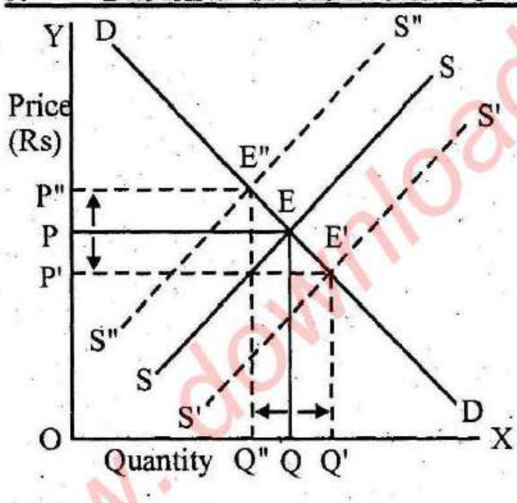
(ii) Equ. Quantity increases by Q' and it is now OQ'.

Effects: (b)

(i) Equ. P falls to P'' and now it is OP''

(ii) Equ. Quantity decreases to Q" and becomes OQ''

3. DEMAND UNCHANGED BUT SUPPLY RISES AND FALLS



Explanation:

SS and DD meet each other at point E. It indicates OP equ. Price and OQ equ. Quantity.

(a) With no change in DD, S'S' moves downward showing a rise in supply with new equ. Point 'E.

Effects:

(i) Price falls from OP to OP'

(ii) Quantity increases from OQ to OQ'.

(b) S''S'' shifts upward when there is a fall in supply. E" becomes new equ. Point.

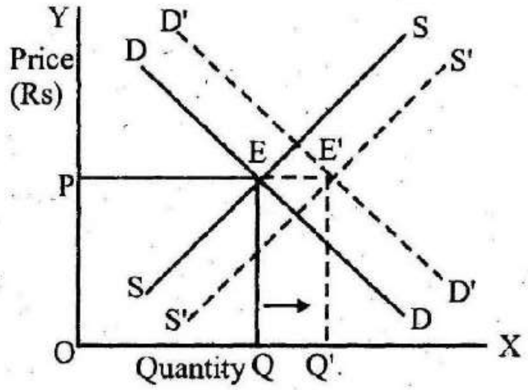
Effects:

(i) Price rises from OP to OP".

(ii) Quantity decreases from OQ to OQ".

4. EQUAL RISE IN DEMAND AND SUPPLY.

(Movement in the same direction)



Explanation

E being original equ. Point shows op as equ. P and OQ as equ. Quantity. With the rise in demand D'D' shifts upward while, an equal rise in supply shows downward shift of (Ss as S'S'. E' becomes new equ. Point).

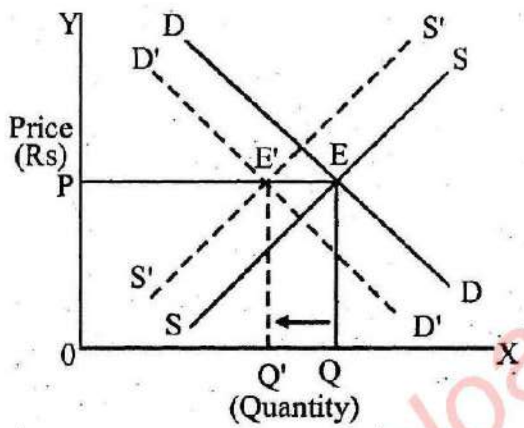
Effects

(i) Price remains the same.

(ii) Quantity rises from OQ to OQ'.

5. EQUAL FALL IN DEMAND AND SUPPLY

(Simultaneous movement in the same direction)



Explanation:

DD and SS at point E indicate equ. Price op and equ. Quantity OQ.

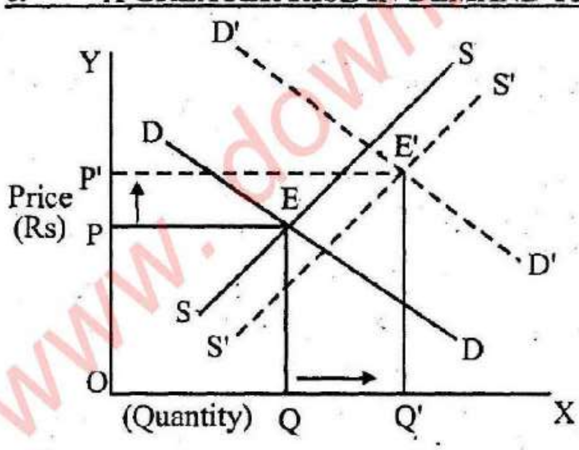
With simultaneous equal fall in Demand and Supply. D'D' shifts downward and S'S' shifts upward showing E' as new equilibrium point.

Effects:

(i) Price remains unchanged.

(ii) Quantity decreases from OQ to OQ'.

6. A GREATER RISE IN DEMAND THAN SUPPLY.



Explanation:

At point E equ. Price is op and equilibrium quantity OQ.

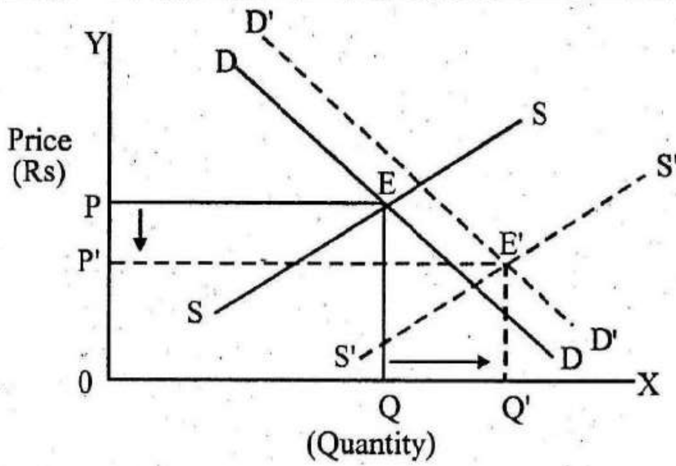
With more rise in demand D'D' shifts upward showing a large distance from DD While, due to comparatively a little rise in supply S'S' shifts downward showing E' as a new equ. Point.

Effects:

(i) Price rises from op to op'.

(ii) Quantity rises from OQ to OQ'.

7. A SMALLER RISE IN DEMAND THAN SUPPLY.



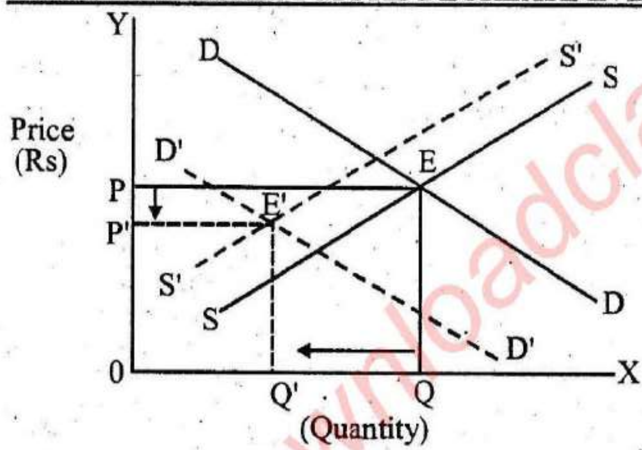
Explanation:

Equ. Point E indicates op equ price and OQ equ. Quantity. Due to a little increase in demand D'D' moves upward showing smaller distance from DD. On the other hand S'S' due to comparatively greater rise in supply shows a large downward movement from SS. 'E becomes the new equ. Point.

Effects:

(i) Price falls from op to op'. (ii) Quantity rises from OQ to OQ'.

8. COMPARATIVELY MORE DECREASE IN DEMAND THAN FALL IN SUPPLY.



Explanation:

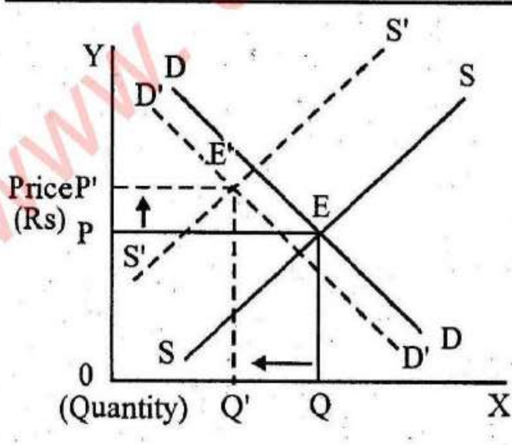
E is the original equ. Point op is equ. Price and OQ equ. Quantity. Due to more decrease in demand D'D' shifts downward to left.

On the other hand S'S' shifts upward to right due to a smaller decrease in supply. E' becomes new equ. Point.

Effects:

(i) Price falls from op to op'. (ii) Quantity decreases from OQ to OQ'.

9. A LITTLE FALL IN DEMAND THAN DECREASE IN SUPPLY.



Explanation:

At point E equ. Price is op and equ. Quantity OQ.

Due to a smaller fall in demand D'D' moves downward nearer to DD. But due to more decrease in supply S'S' shifts upwardly keeping a clear distance from SS. E' is the new equ. Point.

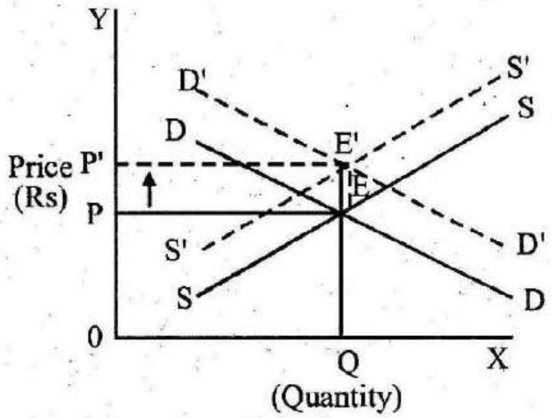
Effects:

(i) Price rises from op to op'.

(ii) Quantity decreases from OQ to OQ'.

10. INCREASE IN DEMAND BUT DECREASE IN SUPPLY.

(Equal change in opposite direction)



Explanation:

At point E equ. price is op and equ. quantity OQ.

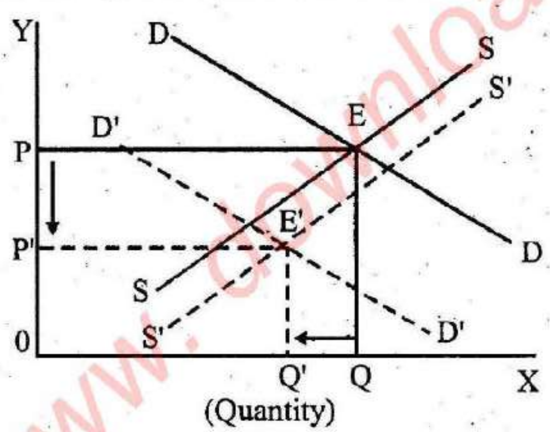
Due to increase in demand D'D' shifts upward to DD. An equal decrease in supply shifts S'S' upwardly showing E' as new equ. point.

Effects:

(i) Price rises from op to op'.

(ii) Quantity remains unchanged

11. MORE FALL IN DEMAND BUT LESS RISE IN SUPPLY.



Explanation:

Equilibrium point E shows that op is equ. price and OQ equ. quantity.

Due to more fall in demand D'D' shifts downward sufficiently away from DD.

As there is a little rise in supply thus S'S' shifts downward to SS with a small distance.

E' is the new equilibrium point.

Effects:

(i) Price rises from op to op'.

(ii) Quantity decreases from OQ to OQ'.

OBJECTIVE QUESTIONS

Select and tick (✓) Correct answer.

1. Equilibrium price is determined where:
   (a) Profit of sellers is maximum
   (b) Determined by the Govt.
   (c) Quantity supplied = Quantity demanded
   (d) According to choice of consumers.
2. If increase in demand is more than increase in supply then:
   (a) Price falls (b) Price rises
   (c) Quantity decreases (d) No change in price and quantity
3. When increase in demand = decrease in supply. What will be the effect upon equilibrium quantity?
   (a) It will rise (b) It will decrease
   (c) No change (d) Rising and falling trend.
4. If, demand and supply decrease equally. How equ. Price will be affected.
   (a) Increase in price (b) Decrease in price
   (c) No change in price (d) Price will fluctuate
5. Find equ. Price and equ. Quantity in view of following equations (i) $qd = 10 - 2p$ (ii) $qs = 4 + p$

	Equ. P	Equ. Q
(a)	1	10
(b)	2	6
(c)	3	4

6. Show the direction of the movement of demand and supply curves.
   (a) Same direction (b) Opposite direction
   (c) Parallel to each other (d) Horizontal position
7. What will be the effect upon equ. Quantity of decrease in demand = increase in supply.
   (a) Will increase (b) Will decrease
   (c) No change (d) Dis-equilibrium.
8. When increase in demand and supply is equal equ. Price will show.
   (a) No change (b) An increase (c) Decrease (d) p = o
9. When demand decreases at fixed supply equ. Price will show.
   (a) No change (b) Increase
   (c) Decrease (d) Control upon price
10. When demand = supply under perfect competition. What will be the reaction?
   (a) A decrease in output (b) Increase in demand by consumers
   (c) Equilibrium price determination
   (d) Dis-equilibrium in the market.

DESCRIPTIVE PART – I

SHORT NOTES

1. What is "Market equilibrium."
2. How equ. Price and quantity will be affected with the decrease in demand at fixed supply.
3. Find out equ. Price and quantity w.r.t. following equations.
   (i) $qd = 25 - 2p$(ii) $qs = 10 + 4p$
4. What will be the effect upon price and quantity with an increase in demand when supply remains fixed.
5. Find equ. Price and quantity with the given demand and supply equations.
   (i) $qd = 20 - 5p$(ii) $qs = 4 + 3p$

PART – II

1. Explain demand and supply equilibrium with the help of schedule and diagram.
2. Bring out the effects of following upon equ. Price and equ. Quantity.
   (a) Demand and supply both rise, but relative rise in demand is greater than supply.
   (b) Equal decrease in demand and supply.
   (c) Change in the demand of a perishable good.
   (d) Increase in demand but decrease in supply (equal change)
3. Solve the following demand and supply equations, to find out equ. Price and quantity. Explain with the help of gaphs.

(a) $qd = 28 - 4p$, $qs = 16 + 2p$

(b) $qd = 25 - 3p$, $qs = 10 + 4p$

(c) $q + p = 5$, $2p - q = 5.5$

(d) $q - 128 + 9p = 0$, $q + 32 - 7p = 0$

(e) $qd = 20 - 5p$, $qs = 4 + 3p$

Ans:

(a) equ. $P = 2$ equ. $P = 20$ (b) equ.$p = 5$, equ. $q = 10$

(c) equ. $p = 3.5$,equ. $q=1.5$ (d) equ. $p = 10$, equ. $q = 38$

(e) equ. $p = 2$, equ. $q = 10$

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